📑 **Technical Report**

Project Title: Customer Churn Analysis and Business Recommendations

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1. Introduction

Customer churn is a critical challenge for subscription-based businesses. High churn rates reduce revenue stability and increase acquisition costs. This project aims to analyze churn drivers using a dataset of 500 customers, apply advanced analytics, and provide actionable recommendations to reduce churn.

Objectives:

- Identify key factors influencing churn.
- Build predictive models to classify churn risk.
- Segment customers into meaningful groups.
- Validate statistical relationships between churn and customer attributes.
- Provide business recommendations with measurable impact.

Success Metrics:

- Predictive accuracy of churn classification.
- Identification of high-risk customer segments.
- Statistically significant insights ($p < 0.05$).
- Practical recommendations aligned with business goals.

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2. Data Collection & Preparation

Dataset: 500 rows, 9 columns.
Features:

- CustomerID
- Tenure (months)
- MonthlyCharges
- TotalCharges
- Contract (Month-to-Month, One Year, Two Year)

- PaymentMethod
- PaperlessBilling
- SeniorCitizen (0/1)
- Churn (Yes/No)

Cleaning Steps:

- Removed duplicates.
- Standardized column names.
- Converted numeric fields (Tenure, MonthlyCharges, TotalCharges).
- Encoded categorical variables.
- Normalized Churn values (Yes/No → 1/0).

Final Dataset: 500 rows, 20+ features after encoding.

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3. Exploratory Data Analysis (EDA)

Key Visualizations:

- **Churn Distribution:** ~26% of customers churned.
- **Tenure vs Churn:** Short-tenure customers churn more frequently.
- **Monthly Charges vs Churn:** Higher charges correlate with churn.
- **Contract Type vs Churn:** Month-to-month contracts show highest churn.
- **Correlation Heatmap:** Strong correlation between TotalCharges and Tenure.

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4. Advanced Analysis

4.1 Logistic Regression (Classification)

- **Model Accuracy:** ~80%
- **Precision/Recall:** Balanced performance, good at identifying churners.
- **Confusion Matrix:** Correctly classified majority of churn cases.

4.2 KMeans Clustering (Segmentation)

- **Clusters Identified:**
    1. **High Risk:** Short tenure, high monthly charges.

    2. **Moderate Risk:** Medium tenure, moderate charges.
    3. **Low Risk:** Long tenure, stable charges.
- **Business Insight:** Segment 1 requires immediate retention strategies.

4.3 Hypothesis Testing

- **Test:** Monthly charges difference between churned vs retained customers.
- **Result:** p-value < 0.05 → statistically significant.
- **Conclusion:** Higher monthly charges increase churn likelihood.

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5. Insights & Recommendations

Insights:

- Short-tenure, high-charge customers are most likely to churn.
- Month-to-month contracts are unstable compared to long-term contracts.
- Payment method and paperless billing show secondary influence.

Recommendations:

1. **Retention Campaigns:** Target Segment 1 with discounts or loyalty rewards.
2. **Contract Incentives:** Encourage long-term contracts with bundled offers.
3. **Pricing Strategy:** Review high monthly charges; offer tiered pricing.
4. **Predictive Monitoring:** Deploy churn prediction model to flag at-risk customers.

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6. Implementation Plan

- **Timeline:** 3 months rollout.
- **Resources:** Marketing team, data science team, CRM integration.
- **KPIs:**
    - Reduce churn by 10–15%.
    - Increase average tenure by 6 months.
    - Improve customer lifetime value.

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7. Conclusion

This analysis demonstrates that churn is driven primarily by contract type, tenure, and monthly charges. By implementing targeted retention strategies and leveraging predictive analytics, the company can significantly reduce churn, stabilize revenue, and strengthen customer loyalty.